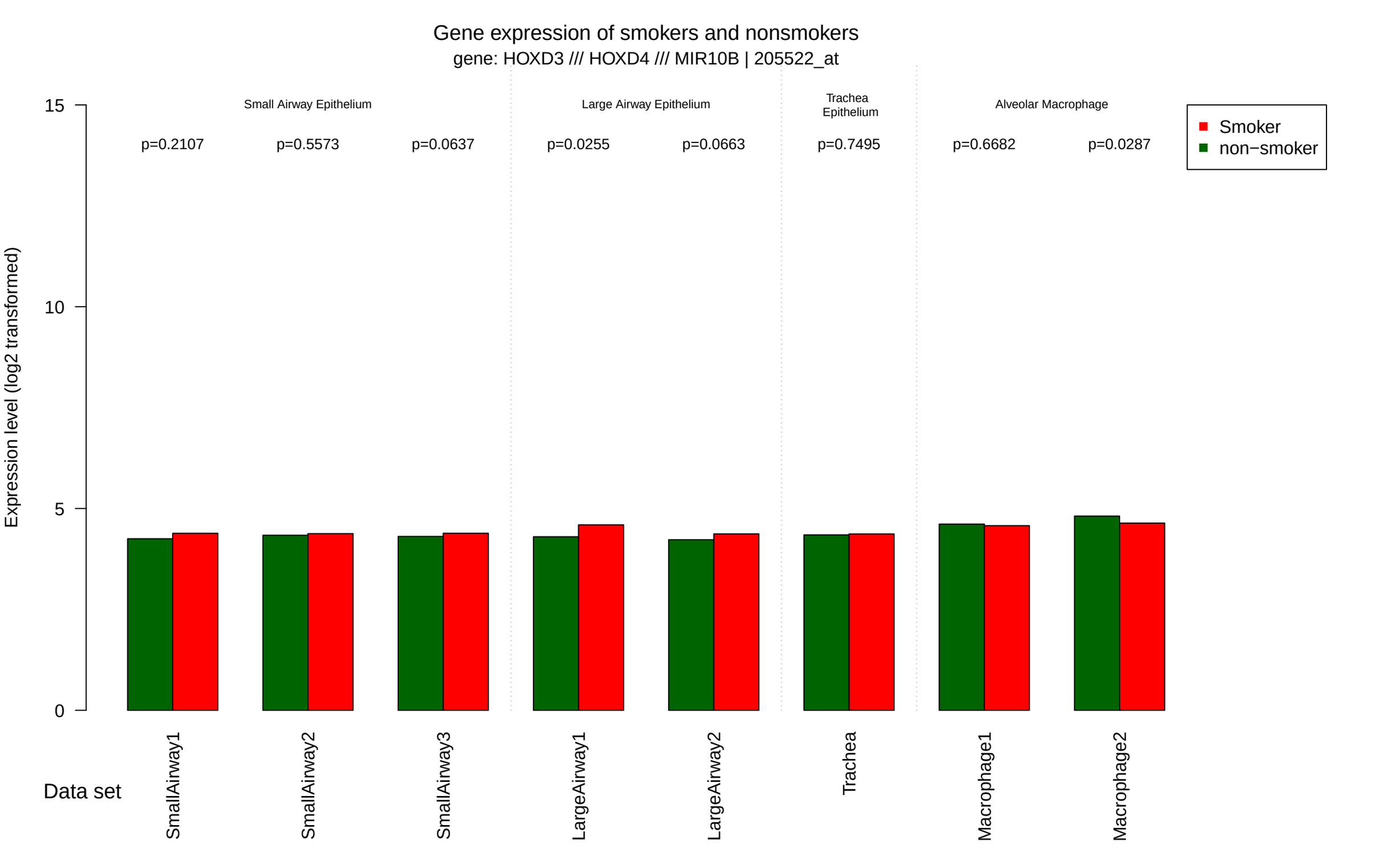
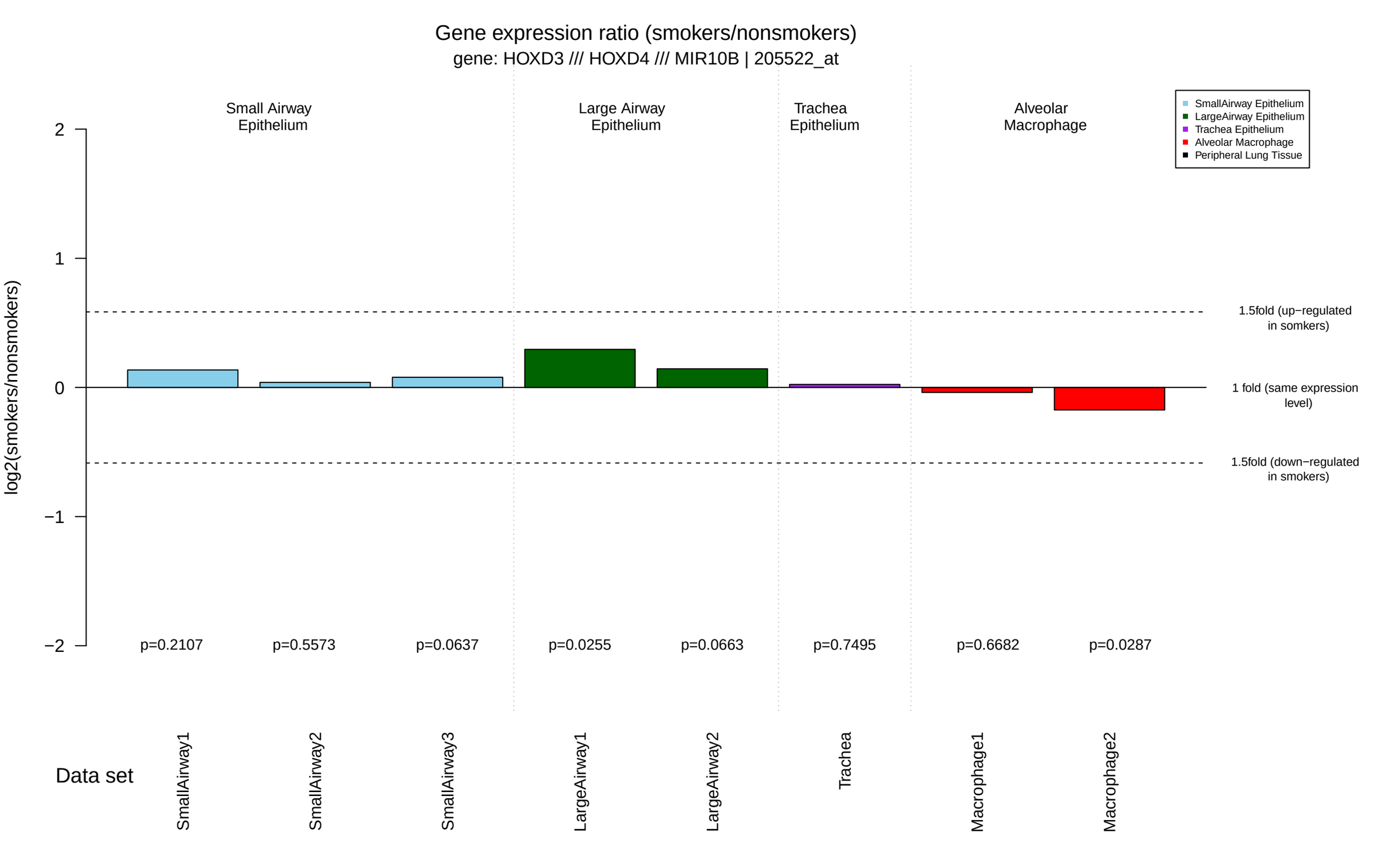
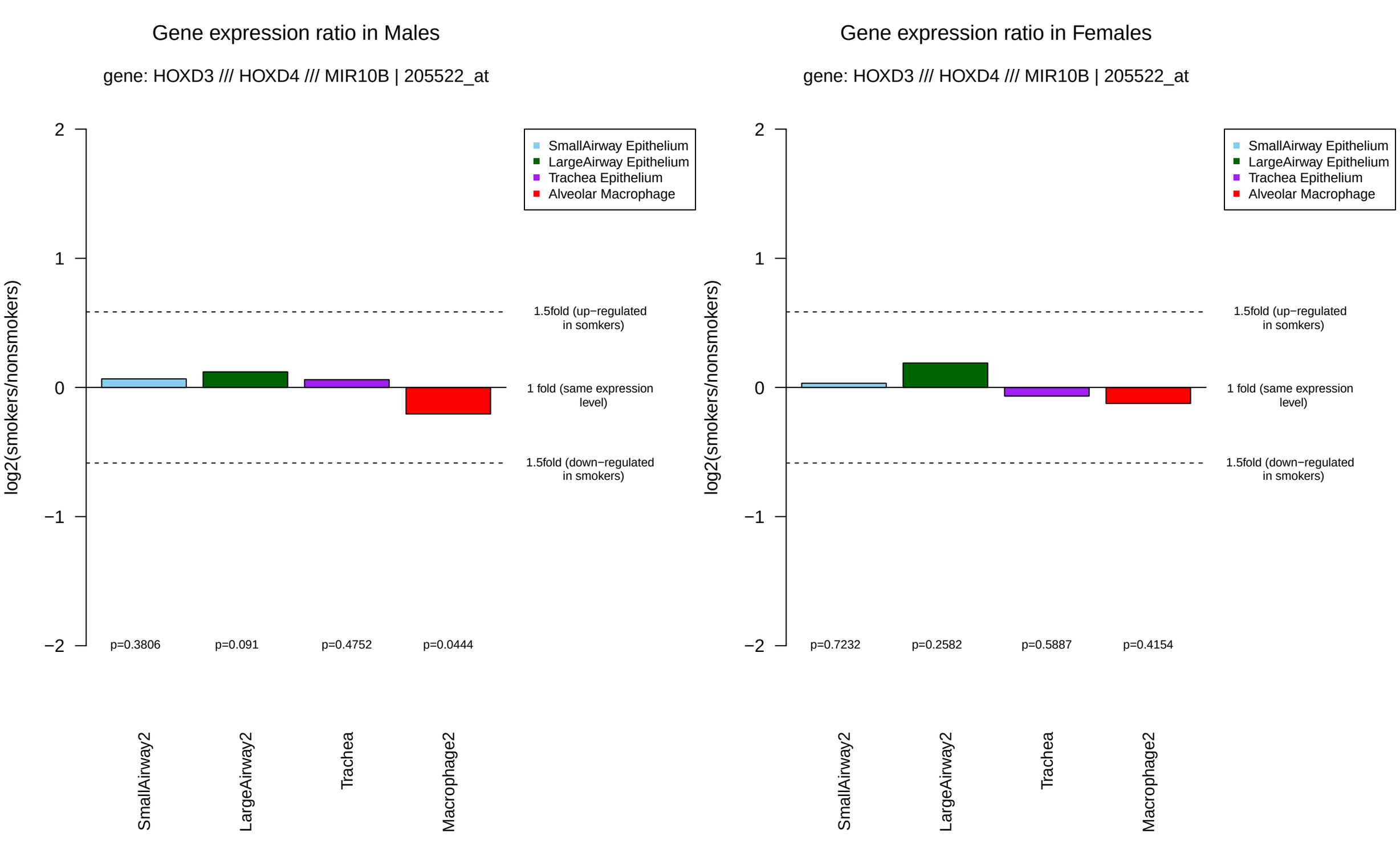
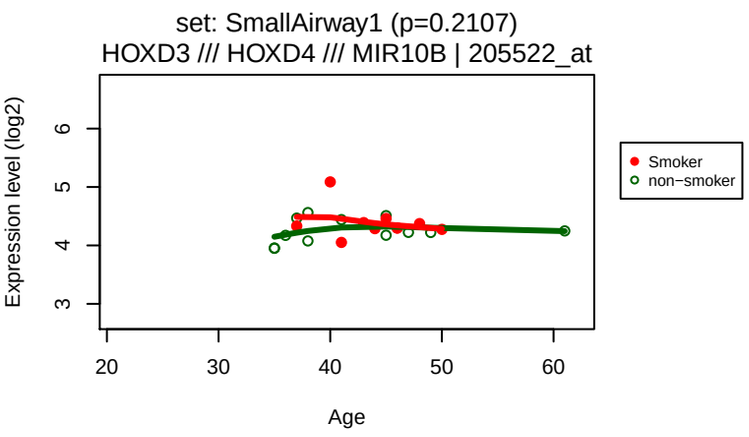
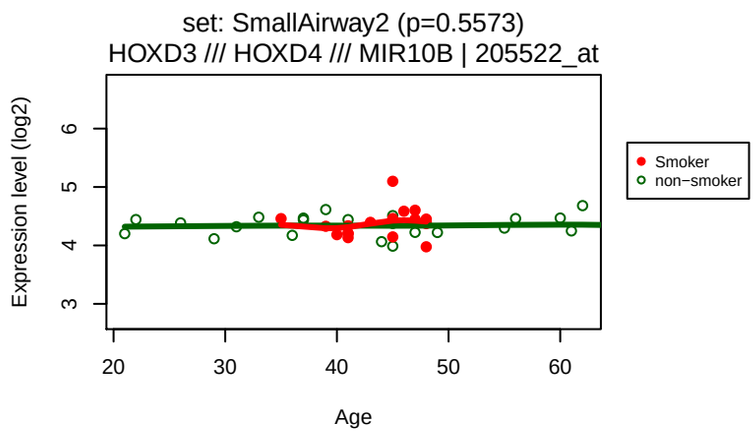
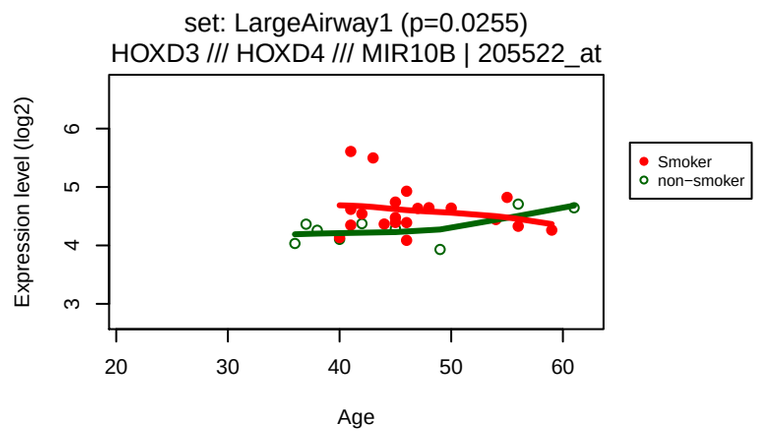

Gene expression of smokers and nonsmokers
gene: HOXD3 /// HOXD4 /// MIR10B | 205522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2107
p=0.5573
p=0.0637
p=0.0255
p=0.0663
p=0.7495
p=0.6682
p=0.0287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

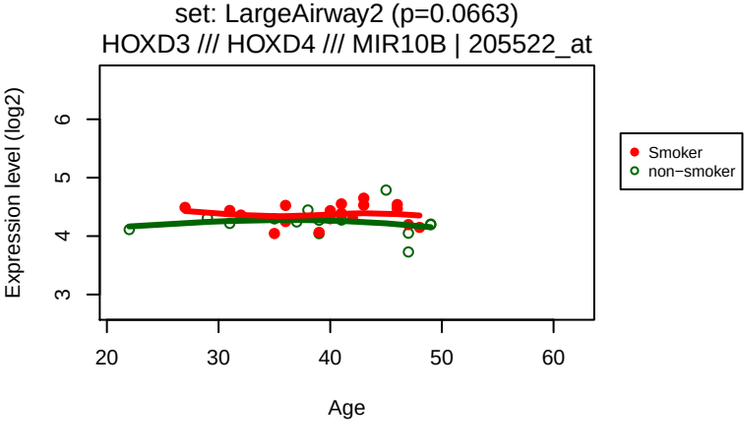

Gene expression ratio (smokers/nonsmokers)
gene: HOXD3 /// HOXD4 /// MIR10B | 205522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2107
p=0.5573
p=0.0637
p=0.0255
p=0.0663
p=0.7495
p=0.6682
p=0.0287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

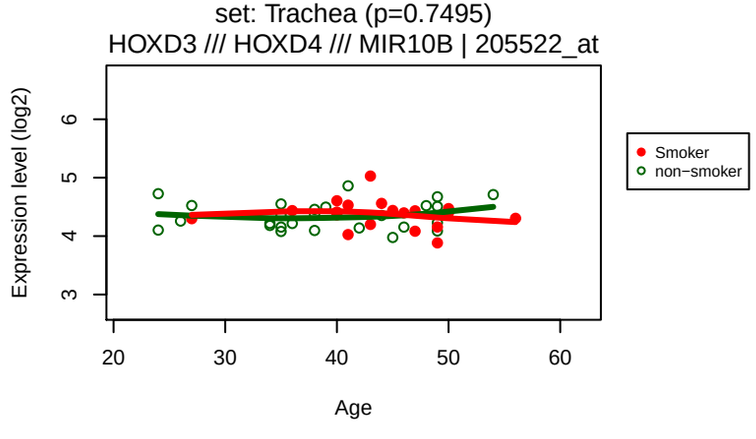

Gene expression ratio in Males
gene: HOXD3 /// HOXD4 /// MIR10B | 205522_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3806
p=0.091
p=0.4752
p=0.0444
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD3 /// HOXD4 /// MIR10B | 205522_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7232
p=0.2582
p=0.5887
p=0.4154
SmallAirway2
LargeAirway2
Trachea
Macrophage2

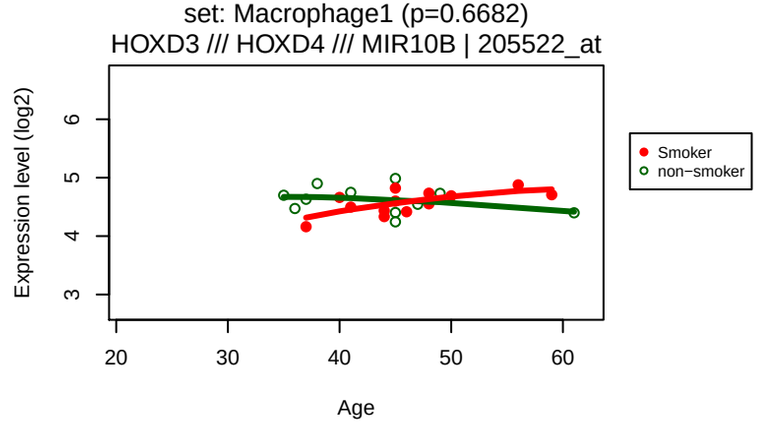

set: SmallAirway1 (p=0.2107)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker

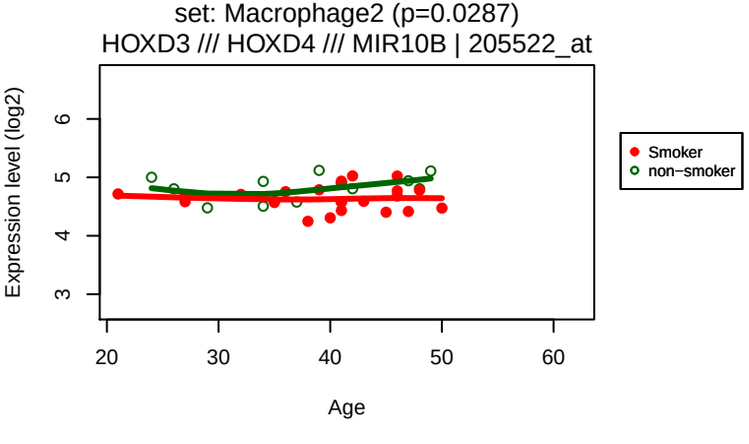

set: SmallAirway2 (p=0.5573)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker

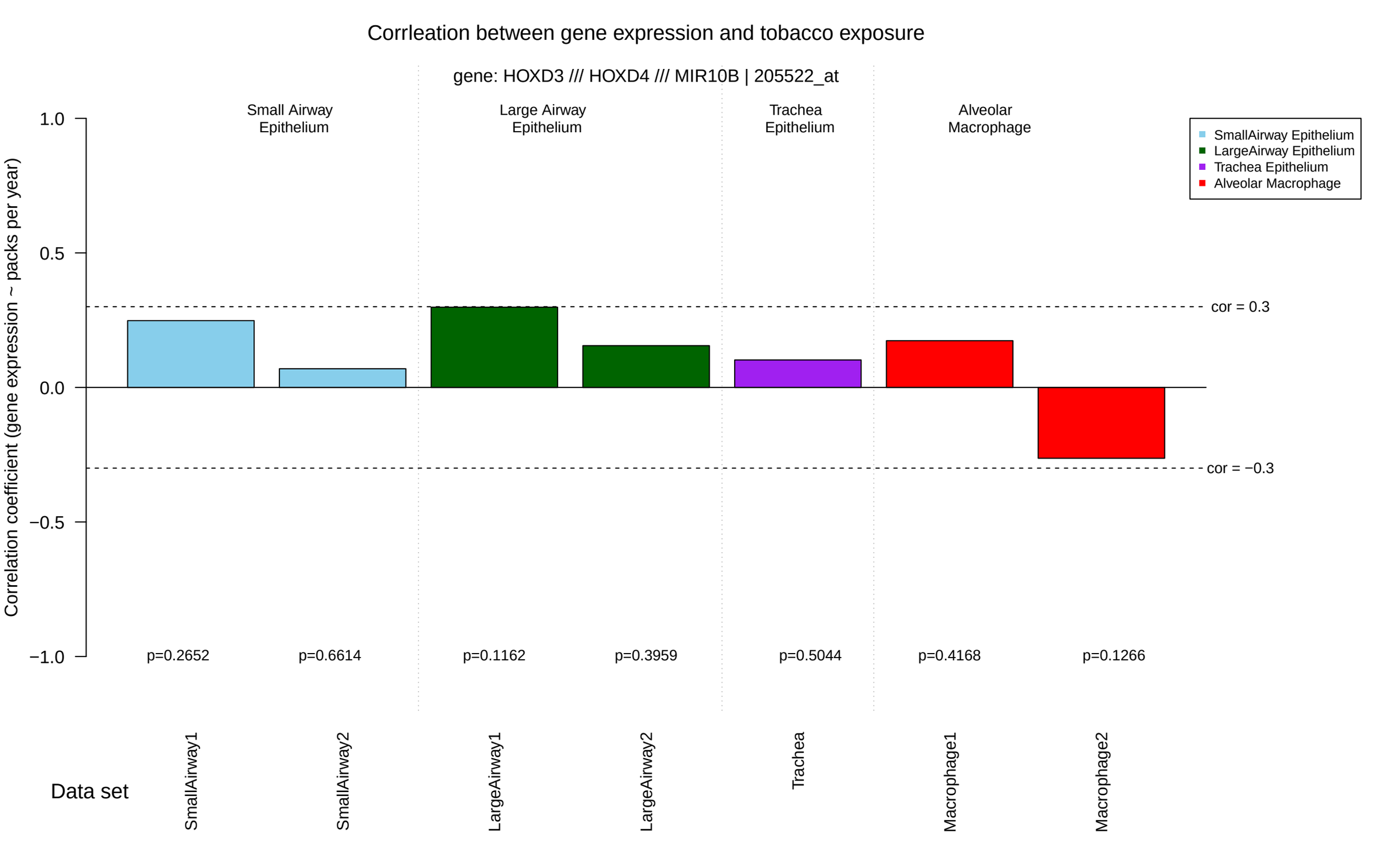

set: LargeAirway1 (p=0.0255)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0663)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7495)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6682)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0287)
HOXD3 /// HOXD4 /// MIR10B | 205522_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD3 /// HOXD4 /// MIR10B | 205522_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2652
p=0.6614
p=0.1162
p=0.3959
p=0.5044
p=0.4168
p=0.1266
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2